Rates of Chemical Reactions

The speed at which a chemical reaction happens is called the _____ of _____.

Examples of quick chemical reactions are _____ and combustion (burning).

Examples of slow chemical reactions are _____ and ripening.

Factors that affect the rate of reactions are:

T_____.

C_____ of the reactions.

Surface _____.

Agitation(_____ and stirring)

C_____.

Temperature.

Normally increasing the temperature will _____ the reaction rate.

Concentration

The term concentration refers to the _____ of reactants present in a _____ volume of liquid or gas. Increasing the concentration of reactants will _____ the rate of reaction.

Agitation.

Agitating makes sure that the reactant particles are kept in _____. Agitation _____ the rate of reaction.

Surface area.

Dividing up solid _____ into _____ pieces creates a much larger _____ in contact between the _____. Increasing _____ _____ increases the reaction rate.

Catalysts.

Catalysts are chemicals that _____ up the rate of _____ but are not consumed (_____ up) during the reaction. They can be considered as _____ helpers. Catalysts can work in two ways:

1. They _____ the amount of energy that is _____ to change the _____ into _____.
2. They make it _____ for _____ molecules to _____ and form _____.

Read the article below and use the information to complete the table below.

A **catalyst** is a substance that speeds up the rate of a chemical reaction but is unconsumed (not used up) by the reaction. A catalyst participates(takes part) in the reaction but is neither a reactant or product. Catalysts enable(help) reactions to occur much faster or at lower temperatures because of changes that they induce(cause) in the reactants. Catalysts provide an alternative(different) pathway, one with a lower activation energy, for a reaction to proceed. Molecules that would not have had the energy to react or that

have such low energies that it is likely that they would take a long time to do so are able to react in the presence of a catalyst.

Word	**Meaning**
Unconsumed	
Participates	
Enable	
Alternative	

Enzymes.

Enzymes are _____ catalysts. Many enzymes work in your _____.